Review Questions

The answers to the chapter review questions can be found in the Appendix.

1. Which statements about the `final` modifier are correct? (Choose all that apply.)
   - **A.** Instance and `static` variables can be marked `final`.
   - **B.** A variable is effectively final only if it is marked `final`.
   - **C.** An object that is marked `final` cannot be modified.
   - **D.** Local variables cannot be declared with type `var` and the `final` modifier.
   - **E.** A primitive that is marked `final` cannot be modified.

2. Which of the following can fill in the blank in this code to make it compile? (Choose all that apply.)

```
public class Ant {
   _____ void method() {}
}
```

   - **A.** `default`
   - **B.** `final`
   - **C.** `private`
   - **D.** `Public`
   - **E.** `String`
   - **F.** `zzz:`

3. Which of the following methods compile? (Choose all that apply.)
   - **A.** `final static void rain() {}`
   - **B.** `public final int void snow() {}`
   - **C.** `private void int hail() {}`
   - **D.** `static final void sleet() {}`
   - **E.** `void final ice() {}`
   - **F.** `void public slush() {}`

4. Which of the following can fill in the blank and allow the code to compile? (Choose all that apply.)

```
final ________ song = 6;
```

   - **A.** `int`
   - **B.** `Integer`
   - **C.** `long`
   - **D.** `Long`
   - **E.** `double`
   - **F.** `Double`

5. Which of the following methods compile? (Choose all that apply.)
   - **A.** `public void january() { return; }`
   - **B.** `public int february() { return null;}`
   - **C.** `public void march() {}`
   - **D.** `public int april() { return 9;}`
   - **E.** `public int may() { return 9.0;}`
   - **F.** `public int june() { return;}`

6. Which of the following methods compile? (Choose all that apply.)
   - **A.** `public void violin(int... nums) {}`
   - **B.** `public void viola(String values, int... nums) {}`
   - **C.** `public void cello(int... nums, String values) {}`
   - **D.** `public void bass(String... values, int... nums) {}`
   - **E.** `public void flute(String[] values, ...int nums) {}`
   - **F.** `public void oboe(String[] values, int[] nums) {}`

7. Given the following method, which of the method calls return 2? (Choose all that apply.)

```
public int juggle(boolean b, boolean... b2) {
   return b2.length;
}
```

   - **A.** `juggle();`
   - **B.** `juggle(true);`
   - **C.** `juggle(true, true);`
   - **D.** `juggle(true, true, true);`
   - **E.** `juggle(true, {true, true});`
   - **F.** `juggle(true, new boolean[2]);`

8. Which of the following statements is correct?
   - **A.** Package access is more lenient than `protected` access.
   - **B.** A `public` class that has private fields and package methods is not visible to classes outside the package.
   - **C.** You can use access modifiers so only some of the classes in a package see a particular package class.
   - **D.** You can use access modifiers to allow access to all methods and not any instance variables.
   - **E.** You can use access modifiers to restrict access to all classes that begin with the word `Test`.

9. Given the following class definitions, which lines in the `main()` method generate a compiler error? (Choose all that apply.)

```
// Classroom.java
package my.school;
public class Classroom {
   private int roomNumber;
   protected static String teacherName;
   static int globalKey = 54321;
   public static int floor = 3;
   Classroom(int r, String t) {
      roomNumber = r;
      teacherName = t; } }

// School.java
1: package my.city;
2: import my.school.*;
3: public class School {
4:    public static void main(String[] args) {
5:       System.out.println(Classroom.globalKey);
6:       Classroom room = new Classroom(101, "Mrs. Anderson");
7:       System.out.println(room.roomNumber);
8:       System.out.println(Classroom.floor);
9:       System.out.println(Classroom.teacherName); } }
```

   A. None: the code compiles fine.
   B. Line 5
   C. Line 6
   D. Line 7
   E. Line 8
   F. Line 9

10. What is the output of executing the `Chimp` program?

```
// Rope.java
package rope;
public class Rope {
   public static int LENGTH = 5;
   static {
      LENGTH = 10;
   }
   public static void swing() {
```

```
      System.out.print("swing ");
   } }
```

```
// Chimp.java
1: import rope.*;
2: import static rope.Rope.*;
3: public class Chimp {
4:    public static void main(String[] args) {
5:       Rope.swing();
6:       new Rope().swing();
7:       System.out.println(LENGTH);
8:    } }
```

- **A.** `swing swing 5`
- **B.** `swing swing 10`
- **C.** Compiler error on line 2 of `Chimp`
- **D.** Compiler error on line 5 of `Chimp`
- **E.** Compiler error on line 6 of `Chimp`
- **F.** Compiler error on line 7 of `Chimp`

11. Which statements are true of the following code? (Choose all that apply.)

```
public class Rope {
   public static void swing() {
      System.out.print("swing");
   }
   public void climb() {
      System.out.println("climb");
   }
   public static void play() {
      swing();
      climb();
   }
   public static void main(String[] args) {
      Rope rope = new Rope();
      rope.play();
      Rope rope2 = null;
      System.out.print("-");
      rope2.play();
   } }
```

- **A.** The code compiles as is.
- **B.** There is exactly one compiler error in the code.
- **C.** There are exactly two compiler errors in the code.
- **D.** If the line(s) with compiler errors are removed, the output is `swing-climb`.
- **E.** If the line(s) with compiler errors are removed, the output is `swing-swing`.
- **F.** If the line(s) with compile errors are removed, the code throws a `NullPointerException`.

12. How many variables in the following method are effectively final?

```
public void feed() {
   int monkey = 0;
   if(monkey > 0) {
      var giraffe = monkey++;
      String name;
      name = "geoffrey";
   }
   String name = "milly";
   var food = 10;
   while(monkey <= 10) {
      food = 0;
   }
   name = null;
}
```

- **A.** 1
- **B.** 2
- **C.** 3
- **D.** 4
- **E.** 5
- **F.** None of the above. The code does not compile.

13. What is the output of the following code?

```
// RopeSwing.java
import rope.*;
import static rope.Rope.*;
public class RopeSwing {
   private static Rope rope1 = new Rope();
   private static Rope rope2 = new Rope();
   {
      System.out.println(rope1.length);
   }
```

```
   public static void main(String[] args) {
      rope1.length = 2;
      rope2.length = 8;
      System.out.println(rope1.length);
   }
}
```

```
// Rope.java
package rope;
public class Rope {
   public static int length = 0;
}
```

- **A.** `02`
- **B.** `08`
- **C.** `2`
- **D.** `8`
- **E.** The code does not compile.
- **F.** An exception is thrown.

14. How many lines in the following code have compiler errors?

```
public class RopeSwing {
   private static final String leftRope;
   private static final String rightRope;
   private static final String bench;
   private static final String name = "name";
   static {
      leftRope = "left";
      rightRope = "right";
   }
   static {
      name = "name";
      rightRope = "right";
   }
   public static void main(String[] args) {
      bench = "bench";
   }
}
```

- **A.** 0
- **B.** 1

C. 2

D. 3

E. 4

F. 5

15. Which of the following can replace line 2 to make this code compile? (Choose all that apply.)

```
1: import java.util.*;
2: // INSERT CODE HERE
3: public class Imports {
4:    public void method(ArrayList<String> list) {
5:       sort(list);
6:    }
7: }
```

A. `import static java.util.Collections;`

B. `import static java.util.Collections.*;`

C. `import static java.util.Collections.sort(ArrayList<String>);`

D. `static import java.util.Collections;`

E. `static import java.util.Collections.*;`

F. `static import java.util.Collections.sort(ArrayList<String>);`

16. What is the result of the following statements?

```
public class Test {
   public void print(byte x) {
      System.out.print("byte-");
   }
   public void print(int x) {
      System.out.print("int-");
   }
   public void print(float x) {
      System.out.print("float-");
   }
   public void print(Object x) {
      System.out.print("Object-");
   }
   public static void main(String[] args) {
      Test t = new Test();
      short s = 123;
      t.print(s);
      t.print(true);
```

```
      t.print(6.789);
   }
}
```

- **A.** `byte-float-Object-`
- **B.** `int-float-Object-`
- **C.** `byte-Object-float-`
- **D.** `int-Object-float-`
- **E.** `int-Object-Object-`
- **F.** `byte-Object-Object-`

17. What is the result of the following program?

```
1:  public class Squares {
2:     public static long square(int x) {
3:        var y = x * (long) x;
4:        x = -1;
5:        return y;
6:     }
7:     public static void main(String[] args) {
8:        var value = 9;
9:        var result = square(value);
10:       System.out.println(value);
11:    } }
```

- **A.** `-1`
- **B.** `9`
- **C.** `81`
- **D.** Compiler error on line 9
- **E.** Compiler error on a different line

18. Which of the following are output by the following code? (Choose all that apply.)

```
public class StringBuilders {
   public static StringBuilder work(StringBuilder a,
      StringBuilder b) {
      a = new StringBuilder("a");
      b.append("b");
      return a;
   }
   public static void main(String[] args) {
      var s1 = new StringBuilder("s1");
      var s2 = new StringBuilder("s2");
```

```
      var s3 = work(s1, s2);
      System.out.println("s1 = " + s1);
      System.out.println("s2 = " + s2);
      System.out.println("s3 = " + s3);
   }
}
```

- **A.** `s1 = a`
- **B.** `s1 = s1`
- **C.** `s2 = s2`
- **D.** `s2 = s2b`
- **E.** `s3 = a`
- **F.** The code does not compile.

19. Which of the following will compile when independently inserted in the following code? (Choose all that apply.)

```
1:  public class Order3 {
2:     final String value1 = "red";
3:     static String value2 = "blue";
4:     String value3 = "yellow";
5:     {
6:        // CODE SNIPPET 1
7:     }
8:     static {
9:        // CODE SNIPPET 2
10:    } }
```

- **A.** Insert at line 6: `value1 = "green";`
- **B.** Insert at line 6: `value2 = "purple";`
- **C.** Insert at line 6: `value3 = "orange";`
- **D.** Insert at line 9: `value1 = "magenta";`
- **E.** Insert at line 9: `value2 = "cyan";`
- **F.** Insert at line 9: `value3 = "turquoise";`

20. Which of the following are true about the following code? (Choose all that apply.)

```
public class Run {
   static void execute() {
      System.out.print("1-");
   }
   static void execute(int num) {
      System.out.print("2-");
   }
```

```
    static void execute(Integer num) {
       System.out.print("3-");
    }
    static void execute(Object num) {
       System.out.print("4-");
    }
    static void execute(int... nums) {
       System.out.print("5-");
    }
    public static void main(String[] args) {
       Run.execute(100);
       Run.execute(100L);
    }
}
```

A. The code prints out `2-4-`.

B. The code prints out `3-4-`.

C. The code prints out `4-2-`.

D. The code prints out `4-4-`.

E. The code prints `3-4-` if you remove the method `static void execute(int num)`.

F. The code prints `4-4-` if you remove the method `static void execute(int num)`.

21. Which method signatures are valid overloads of the following method signature? (Choose all that apply.)

```
public void moo(int m, int... n)
```

A. `public void moo(int a, int... b)`

B. `public int moo(char ch)`

C. `public void moooo(int... z)`

D. `private void moo(int... x)`

E. `public void moooo(int y)`

F. `public void moo(int... c, int d)`

G. `public void moo(int... i, int j...)`